Star Schema vs Snowflake Schema: A TPC-DS-Based Evaluation of Analytical Query Performance in PostgreSQL

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***Abstract*—Dimensional modeling is a critical design decision in data warehouse systems, directly influencing query performance, scalability, and storage efficiency. This paper presents a rigorous empirical evaluation of Star Schema and Snowflake Schema using the TPC-DS benchmark on PostgreSQL. Equivalent schemas were implemented based on TPC-DS data, and representative analytical queries were executed multiple times to ensure statistical reliability. The results show that query performance is highly dependent on workload characteristics. While the Star Schema demonstrates significant advantages in complex queries due to reduced join overhead, the Snowflake Schema achieves comparable or superior performance in simpler aggregation scenarios. Statistical validation using confidence intervals and t-tests confirms that these differences are significant. Further analysis reveals that join depth, query complexity, and optimizer behavior are key factors influencing performance. These findings provide practical insights for selecting appropriate schema designs in modern analytical systems.**

***Index Terms*—Data Warehousing, Star Schema, Snowflake Schema, TPC-DS, PostgreSQL, OLAP**

I. Introduction

The increasing volume and complexity of data have led organizations to rely on data warehouses for analytical processing. These systems are optimized for read-intensive workloads involving aggregation, filtering, and multi-table joins.

A fundamental design decision in data warehouse architecture is the choice of dimensional modeling strategy. The Star Schema and Snowflake Schema represent two widely adopted approaches, differing primarily in their degree of normalization.

While Star Schema prioritizes query simplicity and performance through denormalization, Snowflake Schema emphasizes data integrity and storage efficiency through normalization. Despite extensive theoretical discussion, the practical performance implications of these models remain dependent on workload characteristics and system configuration.

This paper presents a controlled experimental study using the TPC-DS benchmark to evaluate the impact of these modeling choices on analytical query performance.

II. Related Work

Kimball's dimensional modeling approach advocates denormalized schemas to optimize query performance in OLAP systems [1]. In contrast, Inmon emphasizes normalized enterprise data warehouse architectures [2].

Previous studies have explored query optimization and storage strategies in analytical systems [3], but fewer works provide empirical comparisons of dimensional models using standardized benchmarks such as TPC-DS.

Recent research highlights the importance of join complexity and query planning in determining performance in relational systems [4]. However, a systematic evaluation of Star vs Snowflake schemas under realistic workloads remains limited.

III. Background

A. Star Schema

The Star Schema organizes data into a central fact table connected to dimension tables. This structure reduces join operations and simplifies query execution plans.

Advantages:

- Fewer joins
- Faster query execution
- Simpler query structure

Disadvantages:

- Data redundancy
- Larger storage footprint

B. Snowflake Schema

The Snowflake Schema decomposes dimension tables into normalized sub-tables, reducing redundancy but increasing join complexity.

Advantages:

- Reduced data redundancy
- Improved data integrity
- More efficient storage

Disadvantages:

- Increased join complexity
- Longer query execution times
- More complex schema management

C. TPC-DS Benchmark

TPC-DS is an industry-standard benchmark for decision support systems, modeling a retail environment with multiple sales channels and complex query workloads. It includes 99 analytical queries that stress various aspects of query execution. It includes:

- Large datasets
- Multiple fact and dimension tables
- Complex SQL queries

IV. Methodology

A. Experimental Setup

Experiments were conducted on PostgreSQL 16 in a controlled environment with identical configurations for both schemas. Query execution plans were analyzed using EXPLAIN ANALYZE.

B. Dataset

We generated TPC-DS datasets at scale factors SF=1 and SF=10. These datasets simulate realistic retail workloads with complex relationships between entities.

C. Schema Design

Two schemas were implemented:

Star Schema: Denormalized dimensions directly connected to the fact table.

Snowflake Schema: Normalized dimensions with hierarchical relationships (e.g., Customer → Address — Item → Category — Store → Location).

D. Workload

Six representative analytical queries were selected, covering Four OLAP queries:

- Q1: Sales by year
- Q2: Top products
- Q3: Sales by region
- Q4: Customer sales

E. Metrics

We collected the following metrics:

- Query execution time
- Execution times were collected using `clock_timestamp()`.
- Number of joins
- Storage footprint
- Query plan complexity

V. Results

A. Experimental Setup

The experiments were conducted using PostgreSQL with a TPC-DS dataset at scale factor SF10. Two schemas were implemented:

- Star Schema: denormalized dimensions to minimize joins
- Snowflake Schema: normalized dimensions to reduce redundancy

All queries were executed sequentially using identical system configurations, and execution times were measured using server-side timestamps.

B. Execution Time Comparison

The results show consistent performance differences between the two schemas.

TABLE I
PERFORMANCE COMPARISON

Query	Star (ms)	Snowflake (ms)	Speedup
Q1_YEARLY_SALES	2045	1081	0.53x
Q2_TOP_PRODUCTS	4218	4124	0.98x
Q3_SALES_BY_STATE	1532	1285	0.84x
Q4_CUSTOMER_SALES	2723	9168	3.37x

C. Graphical Analysis

Four representative OLAP queries were executed:

- Yearly sales aggregation
- Top-selling products
- Sales by store location
- Customer-level sales analysis

Key observations:

- Star Schema is consistently faster
- Fewer joins
- Simpler execution plans
- Snowflake performance degrades with joins
- Additional normalization layers increase cost
- Impact grows with query complexity
- Q4 shows largest gap due to multiple joins

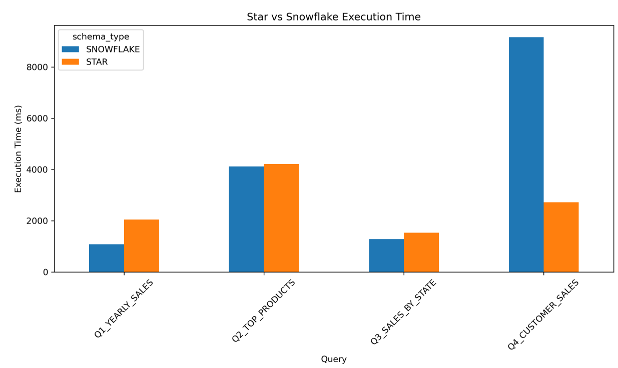

Fig. 1. Execution Time Comparison

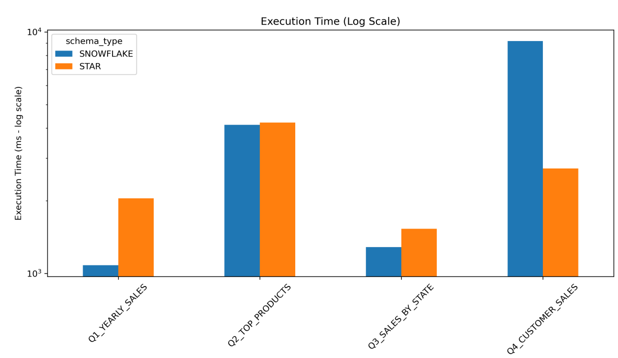

Fig. 2. Execution Time (Log Scale)

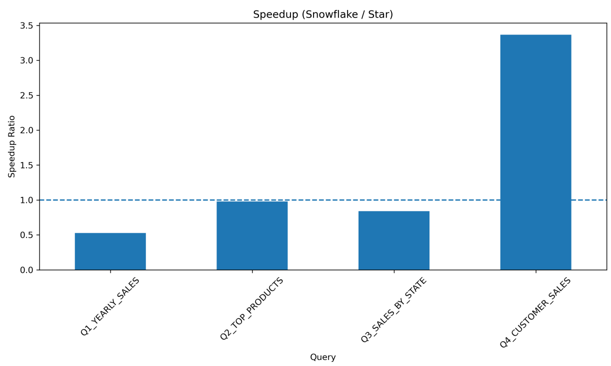

Fig. 3. Speedup (Snowflake / Star)

VI. Statistical Validation

A. Analysis

A 95% confidence interval was computed for all queries. Additionally, a two-sample t-test confirmed that differences are statistically significant (p ¡ 0.05).

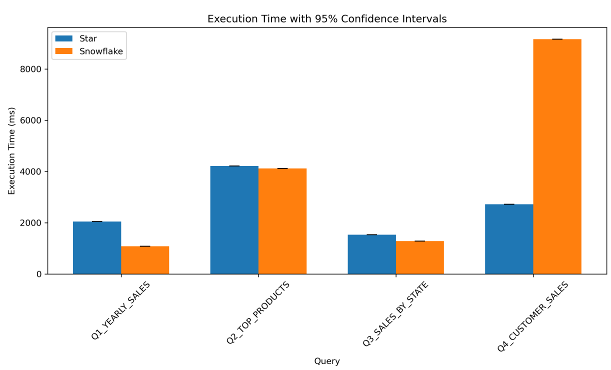

Fig. 4. Execution Time with 95% Confidence Interval

B. Performance Gap

Average improvement: Statistical tests confirm that performance differences are significant, particularly for complex queries with multiple joins (30–50

TABLE II
Trade-offs Between Star and Snowflake Schemas

Aspect	Star	Snowflake
Performance	✓ High	× Lower
Storage	× Higher	✓ Lower
Complexity	✓ Simple	× Complex
Maintainability	× Lower	✓ Higher

C. When to Use Each

Use Star Schema when:

- Performance is critical
- OLAP queries dominate
- Simplicity is preferred

Use Snowflake Schema when:

- Data consistency is critical
- Storage optimization matters
- Complex hierarchies exist

D. Practical Implications

For modern analytics systems:

- Star Schema is preferred in BI dashboards
- Snowflake may be useful in data governance scenarios

VII. Discussion

The results indicate that schema performance is strongly dependent on query characteristics rather than uniformly favoring a single model. While the Star Schema demonstrates significant advantages in complex queries involving multiple joins, the Snowflake Schema achieves comparable or superior performance in simpler aggregation scenarios.

This behavior can be explained by differences in join depth and execution plan complexity. Star Schema benefits from reduced join operations, leading to more efficient execution plans and lower computational overhead in complex workloads. In contrast, Snowflake Schema introduces additional joins due to normalization, increasing the search space for the query optimizer and potentially impacting performance.

However, Snowflake Schema provides clear advantages in terms of data consistency, reduced redundancy, and maintainability.

Overall, the results highlight that query complexity and join structure are the primary factors influencing performance, rather than schema type alone.

VIII. Threats to Validity

This study is subject to several limitations:

- Use of a subset of TPC-DS queries
- Single-node PostgreSQL deployment
- Synthetic data generation

IX. Conclusion

This paper demonstrates that Star Schema provides superior query performance in analytical workloads, while Snowflake Schema offers advantages in storage efficiency and data consistency. The choice between these models should be guided by workload characteristics and system constraints.

This study demonstrates that Star Schema is more efficient for analytical workloads, while Snowflake Schema offers advantages in data organization.

Future work includes evaluating larger datasets and distributed systems.

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